Teaching notes and solutions

> **Teacher note · Teacher notes.** In S21, link each limit to a reading of the graph. In S23, require the signs of the denominator on the left and on the right. In S24, have students choose the transformation before calculating: factorisation, conjugation, local expansion not used. In S26, separate the proof of the comparison of growth rates from its use. Oblique asymptotes are not required here.

Solutions to the activity

1. Since $3x-2 = 3(x-1)+1$, for $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. At $\pm\infty$, $\frac{1}{x-1} \to 0$: the limits equal 3. At 1^-, $x-1 \to 0^-$, so $f(x) \to -\infty$; at 1^+, $x-1 \to 0^+$, so $f(x) \to +\infty$.
3. The sign of $f(x)-3 = \frac{1}{x-1}$ is negative for $x < 1$, positive for $x > 1$. The curve is therefore below and above the asymptote $y = 3$ respectively.
4. For $x > 1$, $|f(x)-3| < 0.01$ is equivalent to $\frac{1}{x-1} < 0.01$, then $x > 101$. We can choose $A = 101$.
5. No: $f(x)-3 = \frac{1}{x-1}$ is never zero. The curve may appear to coincide with the asymptote at the precision of the screen, but it never meets it.

E1. For $x \neq 0$, the quotient equals $\frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. At $+\infty$, the inverse powers tend to 0 and the denominator to $2 \neq 0$. The limit is $\frac{5}{2}$.

E2. At 2, $(x-2)^2 \to 0$ through strictly positive values on both sides; its reciprocal therefore tends to $+\infty$. At $\pm\infty$, the square tends to $+\infty$ and its reciprocal to 0.

E3. For $x \neq 1$, $x^2-1 = (x-1)(x+1)$, so the quotient is equal to $x+1$. Its limit at 1 is therefore 2. This simplification calculates the limit; it does not make the original function defined at 1.

E4. Set $t = 2x \to +\infty$. Then $x^3e^{-2x} = \frac{t^3}{8e^t} \to 0$, by comparison of growth rates of the exponential and the power.

E5. The conjugate gives $\sqrt{x^2+1}+x = \frac{1}{\sqrt{x^2+1}-x}$. As $x \to -\infty$, both terms $\sqrt{x^2+1}$ and $-x$ tend to $+\infty$; the denominator therefore tends to $+\infty$. The limit is 0, through positive values. For $x < 0$, $\sqrt{x^2} = |x| = -x$: the sum $\sqrt{x^2+1}+x$ is therefore of the form $\infty - \infty$, hence the use of the conjugate.

Solutions to Homework 3

1. Since $x^2+1 > 0$ on $\mathbb{R}$, the domain is $\mathbb{R}$. Dividing by x^2 for $x \neq 0$, $f_a(x) = \frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}} \to a$ at $\pm\infty$.
2. $f_a(x)-a = \frac{ax^2+1-a(x^2+1)}{x^2+1} = \frac{1-a}{x^2+1}$. For $a < 1$, the curve is above $y = a$; for $a > 1$, it is below; for $a = 1$, it coincides with this line.

3. By the quotient rule, $f_a'(x) = \frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2} = \frac{2x(a-1)}{(x^2+1)^2}$. If $a > 1$, the sign is that of x: decreasing on $]-\infty, 0]$, increasing on $[0, +\infty[$; minimum $f_a(0) = 1$ and no maximum (the bound a is not attained). If $a < 1$, the variations are reversed: maximum 1 and no minimum. For $a = 1$, the function is constant, with value 1.

4. For $a = 3$ and $x \geq 0$, the inequality becomes $\frac{2}{x^2+1} \leq 0.02$. The denominator is positive, so it is equivalent to $x^2 + 1 \geq 100$, that is $x \geq \sqrt{99}$.

Solutions to the textbook or past-paper work

Exercise 26. As $x \to +\infty$, the functions $f(x) = x$ and $g(x) = x$ both tend to $+\infty$, but their difference is zero. With $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$; swapping f and g, it tends to $-\infty$. These examples establish that the difference is an indeterminate form.

Exercise 29. The denominator vanishes at $\frac{3}{2}$, so $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We write

$$f(x) = \frac{5-2x}{2x-3} = -1 + \frac{2}{2x-3}.$$

Thus $f(x) \to -1$ at both infinities. At $\frac{3}{2}$, the numerator of the second term is positive and its denominator tends to 0: the left-hand limit is $-\infty$, the right-hand limit is $+\infty$. The asymptotes have equations $y = -1$ and $x = \frac{3}{2}$.